

Avinash Mallick

Software Engineer

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Software Engineer

Software engineer with a focus on AI integration and deployment. I work closely with product teams to translate business requirements into scalable, maintainable solutions. My experience includes developing full-stack applications, implementing CI/CD pipelines, and optimizing performance across various technologies.

Technical Skills

Languages: Python, TypeScript, JavaScript, C#, Java, SQL

AI Engineering: LangGraph, LangChain, LlamaIndex, DSPy, RAG, agentic workflows, embeddings, vector search, LLM evaluation, Langfuse, context engineering

Backend & Infra: FastAPI, Pydantic, PostgreSQL, pgvector, Redis, Docker, Caddy, AWS EC2, Cloudflare Pages, GitHub Actions

Enterprise: React, Angular, .NET Core, ASP.NET, Entity Framework, Azure Cloud, IBM Cloud, v-net peering

Experience

Post Urban Ventures

Oct 2025 – Present

AI Systems Engineer - Consultant

London, UK

- Worked directly with founders and product teams to identify operational bottlenecks, scope AI workflows, and ship production-ready prototypes across venture-backed products including Chelo.ai and VFree.
- Built modular AI microservices using FastAPI, Python, LangGraph, PostgreSQL, pgvector, Redis, and Docker, enabling rapid deployment of LLM-powered workflows into startup environments.
- Embedded with insurance brokers to map an unstructured, WhatsApp-based claims workflow, then scoped and shipped a human-in-the-loop agent routing policy questions through retrieval and agent verification (Twilio, production observability) — keeping a broker approval gate on every customer-facing reply.
- Developed RAG pipelines, knowledge graph integrations, and backend orchestration patterns for customer-facing AI products deployed on AWS, Cloudflare, and self-hosted infrastructure.

Converso

Nov 2025 – Present

AI Engineer

London, UK

- Designed and deployed LangGraph-based conversational agents for CRM automation, integrating structured ingestion, retrieval, and customer-specific knowledge workflows for enterprise datasets.
- Deployed a self-hosted Langfuse observability stack, adding tracing, structured logs, latency monitoring, and evaluation workflows to debug and improve agent reliability in production.
- Built document processing and RAG pipelines using embeddings, vector search, validation layers, and human-review patterns to improve trust in AI-generated customer support workflows.
- Partnered with product and engineering stakeholders to translate ambiguous automation requirements into scoped, shipped AI features with monitoring and rollout considerations.

IBM

June 2025 – Sep 2025

Software Engineer, Contract - Joint UCL-IBM Collaboration

London, UK

- Built an agentic code modernization system for Java codebases, using RAG, LangGraph, FastAPI, Pydantic, and IBM Granite models to detect anti-patterns, propose refactors, and generate explainable code changes.
- Designed a multi-agent workflow with scanner, refactor strategist, code transformer, reviewer, and explainer agents, adding compile/test gating to improve reliability of AI-generated refactors.
- Evaluated the system on 10,000+ lines of Java code, achieving 85% anti-pattern detection precision and reducing manual modernization effort by an estimated 60%.

Eurofins Scientific - Eurofins IT Delivery Centre

June 2023 – Sep 2024

Associate Software Engineer

Bangalore, India

- Delivered full-stack eCommerce solutions handling 25,000+ monthly transactions, improving checkout conversion rates by 12% through optimized UI/UX implementations using C#, ASP.NET, and SQL.
- Spearheaded Azure cloud segregation initiative serving 15,000+ users across 8 global regions, reducing deployment time by 45% and infrastructure costs by £30,000 annually.
- Engineered cross-tenant networking solutions using Azure v-net peering, enabling secure data flow for 200+ microservices with 99.8% reliability.

Selected AI Engineering Work

Agentic COBOL to Python Migration

Langflow, Python, Semi-Formal Verification, LLMs

Legacy modernization agent

- Created AI-powered code migration framework achieving 78% automated conversion accuracy on legacy COBOL systems containing 50,000+ lines of code.
- Won DataStax Hacking Agents Hackathon “Best Use Case of Langflow” among 150+ participants from across Europe.

TopoIndex Python Library

Python, RDKit, NetworkX, Poetry, PyPI, GitHub Actions

Molecular analysis package

- Built and published an open-source molecular-analysis library (PyPI, CI via GitHub Actions) for QSAR feature extraction, designed for batch processing of large compound sets in cheminformatics pipelines.
- Implemented validated topological-index algorithms against published reference values, with vectorized graph traversal running ~55% faster than a naïve NetworkX baseline on 1,000+ compound batches.

Achievements

- **Oxford Enspire Guest Lecturer:** Delivered lecture on AI-assisted coding, spec-driven development, and practical AI prototyping for founders at Christ Church College, University of Oxford.
- **Google DeepMind × Cactus Compute Hackathon - 2nd Prize:** Built hybrid local/cloud agentic AI system for privacy-sensitive workflows.
- **AI Tooling in SDLC - Best Use Case of ACI.dev:** Built AI anti-pattern remediation tool using local LLMs, MCP, and GitHub codebase interaction.

Education

University College London

Sept 2024 – Sept 2025

Master of Science, Software Systems Engineering

London, UK

Grade: Merit | Coursework: Engineering of Data Analysis, Applied Deep Learning, Software Development Practices

Hindustan Institute of Technology and Science

July 2019 – April 2023

Bachelor of Technology (Honours), Computer Science and Engineering

Chennai, India

Coursework: Data Structures, Algorithms, Machine Learning, Cloud Computing, Compiler Theory, Distributed Systems